

20/06/2026

Data Base Systems

"Assignment 4"

Question NO. 1:- [Data Base Schema Normalization]

$R = \{ \text{Student_ID}, \text{L_Name}, \text{F_Name}, \text{st_DOB}, \text{st_phone}, \text{st_address}, \text{C_ID}, \text{C_Name}, \text{prof_id}, \text{pre_req}, \text{Grade}, \text{Prof_id}, \text{Prof_LName}, \text{Prof_FName}, \text{Prof_salary}, \text{Prof_phonenumber}, \text{Prof_emailid}, \text{Semester} \}$

Functional Dependencies:

FD1 : $\text{Student_ID} \rightarrow \text{L_Name}, \text{F_Name}, \text{st_DOB}, \text{st_phone}$

FD2 : $\text{st_phone} \rightarrow \text{st_address}$

FD3 : $\text{C_ID}, \text{Student_ID} \rightarrow \text{Grade}$

FD4 : $\text{C_ID} \rightarrow \text{C_Name}, \text{prof_ID}, \text{pre_req}, \text{Prof_emailid}$

FD5 : $\text{Prof_id} \rightarrow \text{Prof_LName}, \text{Prof_FName}, \text{Prof_salary}, \text{Prof_Phonenumber}, \text{Prof_emailid}$

FD6 : $\text{Semester}, \text{C_id} \rightarrow \text{Prof_id}$

$\text{Student_ID}, \text{C_ID}, \text{Semester} \xrightarrow{+} \text{Student_ID}, \text{L_Name}, \text{F_Name}, \text{st_DOB}, \text{st_phone}, \text{st_address}, \text{C_ID}, \text{Semester}, \text{C_Name}, \text{prof_id}, \text{pre_req}, \text{Grade}, \text{Prof_id}, \text{Prof_LName}, \text{Prof_FName}, \text{Prof_salary}, \text{Prof_phonenumber}, \text{Prof_emailid}.$

• Candidate Key = C.K = $\{ \text{Student_ID}, \text{C_ID}, \text{Semester} \}$

♦ Primary Attributes = $\{ \text{Student_ID}, \text{C_ID}, \text{Semester} \}$

♦ Non-Primary Attributes = $\{ \text{L_Name}, \text{F_Name}, \text{st_DOB}, \text{st_phone}, \text{st_address}, \text{C_Name}, \text{prof_id}, \text{pre_req}, \text{Grade}, \text{Prof_LName}, \text{Prof_FName}, \text{Prof_salary}, \text{Prof_phonenumber}, \text{Prof_emailid} \}$

⇒ Currently in 1-NF.

• For 2-NF:

- Partial Dependency Exists.

(Non-Primary Attributes)

• $\swarrow$ L-Name, F-Name, st-DOB, st-phone depending on Proper Subset of CK (Student-ID only).

• Non-Primary Attributes (C-Name, prof-id, pre-req) depending on Proper subset of CK (C-ID only).

• N-PA_s (Grade) depending on Proper Subset of CK (C-ID, Student-ID)

$\Rightarrow$ Decompose to 2-NF:-

• Table 1: Student

Student (Student-ID, L-Name, F-Name, st-DOB, st-phone)

CK = { Student-ID }

FD : Student-ID $\rightarrow$ L-Name, F-Name, st-DOB, st-phone.
st-phone $\rightarrow$ st-address

• Table 2: Course

Course (C-ID, C-Name, prof-id, pre-req)

CK : { C-ID }

FD : C-ID $\rightarrow$ C-Name, prof-id, pre-req.

• Table 3: Professor

Professor (Prof-id, Prof-LName, Prof-FName, Prof-salary,
Prof-Phonenumber, Prof-emailid)

CK : { Prof-id }

FD : Prof-id $\rightarrow$ Prof-LName, Prof-FName, Prof-salary,
Prof-Phonenumber, Prof-emailid.

Duration: 10-15 min ??

• Table 4: Enrollment

Enrollment (Student-ID, C-ID, Semester, Grade)

CK = {Student-ID, C-ID, Semester}

F.D : Student-ID, C-ID $\rightarrow$ Grade

• Table 5: Course_Offering

Course_Offering (C-ID, Semester, Prof-ID)

CK : {C-id, Semester}

F.D : C-ID, Semester $\rightarrow$ Prof-ID

• For 3-NF :

- Transitive Dependencies Exists.

• In Table 1 :

Student-ID $\xrightarrow{\text{NPA}}$ st-phone $\xrightarrow{\text{NPA}}$ st-address [Violation]

• Table 1A: Student

Student (Student-ID, L-Name, F-Name, st-DOB, st-phone)

C.K : {Student-ID}

F.D : Student-ID $\rightarrow$ L-Name, F-Name, st-DOB, st-phone

• Table 1B: Student-Contact

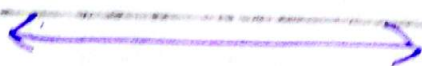
Student-Contact (st-phone, st-address)

C.K : {st-phone}

F.D : st-phone $\rightarrow$ st-address

⇒ Final Normalized Schema (3NF)

- Table 1A
- Table 1B
- Table 2-5.



Question. NO. 2

a) $R(A, B, C, D, E, F, G)$

FD: $\{AC \rightarrow DF, D \rightarrow G, F \rightarrow BE\}$

C.K. $\rightarrow AC \xrightarrow{+} D, F, G, B, E, AC$
 $\hookrightarrow \{AC\}$

PA = $\{A, C\}$

NPA = $\{B, D, E, F, G\}$

- Already in 1-NF
- Also in 2-NF (as there is no Partial Dependency)
- There is Transitive Dependency: (Not in 3-NF)

$AC \rightarrow DF$
 $(\begin{matrix} \text{NPA} & \text{NPA} \\ G & D \end{matrix}) \rightarrow G$
 $\rightarrow F \rightarrow BE$
 $\text{NPA} \quad \text{NPA}$

Highest NF = 2-NF

	$AC \rightarrow DF$	$D \rightarrow G$	$F \rightarrow BE$
1-NF	✓	✓	✓
2-NF	✓	✓	✓
3-NF	✓	X	X

b) $R(\overset{u,v,w,x,y,z}{\cancel{A,B,C,D,E}})$

F.D: $\{u \rightarrow w, uw \rightarrow vx, y \rightarrow ux, y \rightarrow z, uw \rightarrow yz\}$

C.K: $\{u\} + \{y\} + \{uw\}$

1-NF	$u \rightarrow w$	$uw \rightarrow vx$	$y \rightarrow ux$	$y \rightarrow z$	$uw \rightarrow yz$
1-NF	✓	✓	✓	✓	✓
2-NF	✓	✓	✓	✓	✓
3-NF	✓	✓	✓	✓	✓
BCNF	✓	✓	✓	✓	✓

$u^+ \rightarrow wuvxyz$

Highest NF = BCNF

Question 03

Given Table:-

BookID	Title	AuthorName	AuthorDOB	Publisher	publisherAdd
1	DBMS	C.J. Date	1941-12-23	Addison wesley	75 Arlington street, Boston MA
2	DBMS	CJ. Date	1941-12-23	Addison wesley	75 Arlington street, Boston MA
3	DBMS	Thomas H.	1956-10-01	MIT press	one Rogers street MA

~ (i) ~
The table has single valued cell in each field, so its satisfy 1NF.

Functional Dependencies Present:

- BookID $\rightarrow$ Title, publisher, publisher Address
- AuthorName $\rightarrow$ AuthorDOB,
- Publisher $\rightarrow$ publisherAddress.

C.K = { BookID }

Violates 2NF Because

AuthorName $\rightarrow$ AuthorBirthdate

publisher $\rightarrow$ publisherAdd.

Current Normal Form = 1NF

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1NF Already satisfied, all values are single-

2NF - Remove partial dependencies,
But we still have transitive dependencies

3NF - Remove Transitive dependencies.

Relation	Attributes
Book	BookID (PK), Title, Publisher (FK)
Author	AuthorName (PK), - AuthorDOB
BookAuthor	BookID (PK/FK), AuthorName (PK/FK)
Publisher	Publisher (PK), PublisherAddress

Now, no transitive or partial dependencies remain.

All tables in 3NF.

Question 4:-

Order ID	Customer Name	Customer Address	Product ID	Product Name	Quantity
1	John Doe	123 Elm St	101	Laptop	1
2	John Doe	123 Elm St	102	Mouse	2
3	Jane Smith	456 Oak St	103	Keyboard	1

Solution:-

Step 1: Identify Functional dependencies

- $\text{OrderID} \rightarrow \text{CustomerName}, \text{CustomerAddress}$
- $\text{ProductID} \rightarrow \text{ProductName}, \text{UnitPrice}$
- $(\text{OrderID}, \text{ProductID}) \rightarrow \text{Quantity}$

so the composite primary key is:
 $(\text{OrderID}, \text{ProductID})$

Step 2:- Identify partial dependencies

"partial dependency occurs when a non key attribute depends on part of a composite key"

- $\text{CustomerName}, \text{CustomerAddress}$ depends only on OrderID
- $\text{ProductName}, \text{UnitPrice}$ depends only on ProductID

so partial dependencies exist

step 3: convert to 2NF

Removing partial dependencies and splitting table:

orders Table

- orderID (PK)
- CustomerName
- CustomerAddress

products Table

- ProductID (PK)
- ProductName
- UnitPrice

OrderDetails Table

- OrderID (PK, FK)
- ProductID (PK, FK)
- Quantity

step 4: check for 3NF

Check Transitive dependencies

- No attribute depends on another non-key attribute
- All attributes depend only on keys

So the tables are in 3NF

Final 3NF Schema:-

1. orders (orderID, CustomerName, CustomerAddress)
2. products (ProductID, ProductName, UnitPrice)
3. orderDetails (orderID, ProductID, Quantity)

Question 5:-

studentID	studentName	CourseID	Course Name	InsName
1	Alice	101	Math	Dr. Smith
2	Bob	102	History	Dr. Johnson
1	Alice	102	History	Dr. Johnson

1) partial dependencies:-

Composite key is:-

(studentID, CourseID)

Partial dependencies:

- studentID $\rightarrow$ studentName
- CourseID $\rightarrow$ CourseName, InstructorName

So:

- studentName depends only on studentID
- course details depend only on CourseID.

2) Transitive Dependencies:-

As we can see:

- CourseID $\rightarrow$ CourseName
- CourseName $\rightarrow$ InstructorName (bcz instructor is associated with course)

So: studentID, CourseID $\rightarrow$ InstructorName
This creates a transitive dependency.

3) Normalization to 3NF

Decompose the table as follows:

Students Table:-

- StudentID (PK)
- StudentName

Courses Table:-

- CourseID (PK)
- CourseName
- InstructorName

Enrollments Table:-

- StudentID (PK, FK)
- CourseID (PK, FK)

Final 3NF Schema:-

- 1) Students (StudentID, StudentName)
- 2) Courses (CourseID, CourseName, InstructorName)
- 3) Enrollments (StudentID, CourseID)